Subject Description Form

Subject Code	LSGI1101
Subject Title	Spatial Statistics
Credit Value	2
Level	1
Pre-requisite/ **Co-requisite/** **Exclusion**	Nil
Objectives	This subject is to introduce to students the fundamental concepts of spatial statistics, including probability distributions, sampling, and estimation and inference in spatial statistics.
Intended Learning Outcomes *(Note 1)*	Upon completion of the subject, students will be able to: (a) Understand basic statistics, probability, and regression (b) Distinguish different types of spatial data (c) perform basic statistical analyses and inferential procedures on given datasets (d) properly apply spatial statistics techniques to practical problems (e) demonstrate the abilities of logical and analytical thinking
Subject Synopsis/ Indicative Syllabus *(Note 2)*	**Nature of spatial data** Characteristics of spatial data; Special treatment of spatial data in spatial analysis. **Descriptive Statistics** Tables, graphs, frequency distributions, and summary measures of location and dispersion **Introduction to Probability** Experiment, events and probability; Probability rules; Expectation **Discrete Random Variables** Probability density function; Introduction to discrete distributions such as binomial, geometric, and Poisson. **Continuous random variables** Probability density function; Introduction to continuous random variables such as uniform, exponential, normal, etc.; Normal **Sampling Distributions** Population and random samples; Sampling distributions related to sample mean and sample variance **Statistical Inference** Concepts of point estimation and confidence intervals; Point and interval estimation of mean, proportion, variance and difference of

	means or proportions; Tests of significance; Hypothesis tests of mean, proportion, variance and difference of means or proportions. **Regression analysis** Linear regression, non-linear regression, simple regression, multivariable regression, ordinary least square. **Relational spatial data analysis** Basic relational analysis: spatial correlation, proximity, direction; Network analysis; Pattern analysis; Topological analysis.
Teaching/Learning Methodology *(Note 3)*	The subject will be delivered mainly through lectures and tutorials. The lectures will be conducted to introduce the basic statistics concepts of the topics in the syllabus. Lab sessions and assignments to reinforce the theories and methodology introduced during the lectures, so as to enable students to gain deeper understanding of the principles and techniques, to acquire practical problem-solving skills, to become critical in thinking; A group project is designed to enhance the critical thinking, team spirit, problem solving skill, leadership and presentation skill.
Assessment Methods in Alignment with Intended Learning Outcomes *(Note 4)*	(see table and explanation below)

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
		a	b	c	d	e
1. Lab practice	30%	√	√	√	√	√
2. Group project	30%	√	√	√	√	√
3. Written test	40%	√	√	√	√	√
Total	100 %					

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

It consists of 100% continuous assessment through lab practice (30%), group project (30%), and written test (40%). Lab practice includes lab, tutorial and assignments. Through these activities, students will be assessed about the fundamental knowledge in spatial statistics and the practical capabilities of performing spatial data modelling, analysis, and manipulation using actual data sets. Group project is designed to assess the ability of students to apply spatial statistics for solving real-world problems, and project report and presentation will be evaluated. Written test is designed to monitor student learning at knowledge level. Students are expected to achieve a minimum standard to be able to obtain a passing grade in line with criterion referenced assessment approach.

Student Study Effort Expected	Class contact:	
	▪ Lecture	20 Hrs.
	▪ Tutorial	6 Hrs.
	Other student study effort:	
	▪ Assignments	30 Hrs.
	▪ Self-learning	20 Hrs.
	Total student study effort	76 Hrs.
Reading List and References	Mendenhall, W., Beaver, R.J. & Beaver, B.M. Introduction to Probability and Statistics. 15th ed. Thomson (2019) Walpole, R.E., Myers, R.H., Myers, S.L. & Ye, K.Y. Probability and Statistics for Engineers and Scientists. 9th ed. Prentice Hall (2012) Lance A. Waller and Carol A. Gotway Applied Spatial Statistics for Public Health Data. John Wiley & Sons. (2004) Bivand, Pebesma, & Gómez-Rubio. Applied spatial data analysis with R (2013)	